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Expression of functionally glycosylated PSGL-1 in ALK-positive anaplastic large cell lymphoma

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ABSTRACT

Anaplastic lymphoma kinase (ALK)-positive (ALK+) anaplastic large cell lymphoma (ALCL) is defined as a CD30-positive mature T-cell lymphoma characterized by proliferation of large anaplastic cells and aberrant ALK protein expression. These cells often infiltrate lymphatic sinuses, proliferate and exhibit intercellular cohesiveness and accumulate around blood vessels. However, molecular mechanisms underlying such vascular-related histogenesis remain to be elucidated. Since PSGL-1 is more highly expressed in ALK+ ALCL than in other T-cell lymphomas and sLe^x glycan expression has been reported in ALK+ ALCL, the interaction between PSGL-1, which is functionally glycosylated with sLe^x, and P-selectin, which is expressed on endothelial cells, may contribute to such vascular-related histogenesis. To analyze glycans on PSGL-1 expressed in ALK+ ALCL in the context of P-selectin binding function, we first performed immunohistochemical analysis of 12 cases of ALK+ ALCL to identify those that express both PSGL-1 and sLe^x glycans and found that a substantial portion of both molecules co-localizes in those cases. We then conducted western blot analysis, together with glycosyltransferase gene expression analysis, of ALK+ ALCL cells and observed that PSGL-1 is decorated with sLe^x glycans, especially those displayed on core 2 *O*-glycans. Finally, we carried out a P-selectin•IgM chimera binding assay to show that ALK+ ALCL cells bind to P-selectin. These results indicate that the PSGL-1 glycoform expressed on ALK+ ALCL cells is functional and that interaction of functionally glycosylated PSGL-1 with P-selectin may be partially responsible for the vascular-related histogenesis characteristics of ALK+ ALCL.

1. Introduction

Anaplastic lymphoma kinase (ALK)-positive (ALK+) anaplastic large cell lymphoma (ALCL) was first described as "anaplastic large-cell Ki-1-positive lymphoma" by Stein *et al.* in 1985 [1] based on anaplastic morphology and lymphoma cell reactivity to the Ki-1 monoclonal antibody, which was later found to recognize CD30 [2,3]. Based on the latest edition of the World Health Organization (WHO) classification, ALK+ ALCL is defined as CD30-positive mature T-cell lymphoma with aberrant expression of ALK protein secondary to *ALK* gene rearrangements [4]. ALK+ ALCL is relatively rare in adults but is the most common T-cell lymphoma in children and young adults, accounting for 10–30 % of all lymphomas, with a male-to-female ratio of 3:1 [4,5]. Histologically, ALK+ ALCL consists of proliferative large neoplastic cells with abundant cytoplasm and horseshoe-shaped or kidney-like nuclei (called "hallmark" cells) [4]. Interestingly, lymphoma cells often infiltrate lymphatic sinuses and proliferate with intercellular cohesiveness, an activity that resembles metastatic carcinoma. Lymphoma cells also

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often cluster around blood vessels [4]. However, molecular mechanisms underlying such ALK+ ALCL characteristics remain to be elucidated.

P-selectin glycoprotein ligand 1 (PSGL-1) is a 120 kDa mucin-like transmembrane protein expressed as a disulfide-linked homodimer on most leukocytes [6,7]. In humans, PSGL-1 binds to P-selectin expressed on endothelial cells only when i) the threonine residue at position 57 (Thr57) located at the N-terminus of the extracellular domain is modified with a core 2 *O*-glycan capped with sialyl Lewis x (sLe^x) tetrasaccharide, and ii) at least one of the three tyrosine residues clustered in its vicinity (Tyr46, Tyr48 or Tyr51) is sulfated [8,9]. In addition to binding P-selectin, PSGL-1 also reportedly binds to CCL19 and CCL21 chemokines, thereby facilitating immune cell migration to secondary lymphoid organs [10,11]. These molecular interactions are presumed to function in leukocyte trafficking in both physiological conditions and in pathologic conditions, including hematological malignancies.

Recently, PSGL-1 was shown to be more highly expressed in ALK+ ALCL than in other T-cell lymphomas, such as angioimmunoblastic T-cell lymphoma (AITL) and peripheral T-cell lymphoma, not otherwise specified (PTCL, NOS) [12]. Moreover, Kasai *et al.* previously reported expression of sLe^x glycans recognized by the CSLEX-1 monoclonal antibody [13] in all ALK+ ALCL cases examined ($n = 3$) [14]. However, it remains to be determined i) whether PSGL-1 expressed on ALK+ ALCL is decorated with core 2 *O*-glycans terminated with sLe^x tetrasaccharide, and if so, ii) whether that PSGL-1 glycoform is functional, that is, capable of binding to P-selectin.

Here, to answer these questions, we first performed immunohistochemical analysis of 12 ALK+ ALCL cases and two cell lines derived from ALK+ ALCL to determine expression patterns of PSGL-1 and sLe^x glycans in ALK+ ALCL. We next conducted western blot analysis and glycosyltransferase gene expression analysis of ALK+ ALCL cells to determine whether PSGL-1 expressed on ALK+ ALCL cells is decorated with core 2 *O*-glycans terminated with sLe^x tetrasaccharide. Finally, we carried out a P-selectin•IgM chimeric protein binding assay to determine whether the PSGL-1 glycoform expressed on ALK+ ALCL cells functions as a P-selectin ligand. Based on results presented here, we conclude that functionally glycosylated PSGL-1 contributes to vascular-related histogenesis characteristics of ALK+ ALCL.

2. Materials and methods

2.1. Human tissue samples

Formalin-fixed, paraffin-embedded (FFPE) tissue sections from patients diagnosed with ALK+ ALCL ($n = 12$) were obtained from the pathology archive of University of Fukui Hospital and its five affiliated hospitals. Tissue specimens were reviewed by board-certified pathologists (NY and MF), who reconfirmed that the histological and immunohistological appearance of all 12 cases best fit with a diagnosis of ALK+ ALCL (see "Results"). The age of patients who provided samples ranged from 5 to 59 years, with a median of 14.5 years and a male-to-female ratio of 2:1 (Table 1). Analysis of human ALK+ ALCL tissues was approved by the Research Ethics Committee of University of Fukui (reference number 20220093) on September 7, 2022.

2.2. Cell lines

The human ALK+ ALCL cell line Ki-JK [15] was obtained from the Japanese Collection of Research Bioresources Cell Bank (Ibaraki, Japan). Another human T-cell lymphoma line, Karpas 299 [16], was obtained from the European Collection of Authenticated Cell Cultures (Porton Down, UK) through KAC (Amagasaki, Japan). A human T-cell leukemia line, Jurkat, was obtained from the American Type Culture Collection (ATCC; Manassas, VA) and used as a control. These cells were cultured under standard conditions in RPMI 1640 medium supplemented with 10 % fetal bovine serum.

2.3. Monoclonal antibodies

The following monoclonal antibodies served as primary antibodies: Ber-H2 (mouse IgG; Roche Diagnostics, Mannheim, Germany), which recognizes human CD30; D5F3 (rabbit IgG; Roche Diagnostics), which recognizes human ALK; KPL-1 (mouse IgG; Santa Cruz Biotechnology, Dallas, TX), which recognizes human PSGL-1; and HECA-452 (rat IgM; BD Biosciences, Franklin Lakes, NJ), which recognizes sLe^x regardless of 6-*O*-sulfation of the *N*-acetylglucosamine (GlcNAc) residue comprising sLe^x tetrasaccharide [17–19].

2.4. Immunohistochemistry

Immunohistochemical staining for Ber-H2 and D5F3 was performed using a VENTANA BenchMark ULTRA autostainer (Roche Diagnostics). KPL-1 immunostaining was performed using the Histofine system (Nichirei Biosciences, Tokyo, Japan), as described [20,21]. Finally, HECA-452 staining was performed using an indirect method, as described [22,23].

2.5. Double immunofluorescence

FFPE tissue sections and cell suspensions were incubated 60 min with a cocktail of KPL-1 and HECA-452 antibodies. After washing, samples were incubated with a cocktail of DyLight 550-labeled goat anti-mouse IgG (Bethyl Laboratories, Montgomery, TX) for KPL-1 and DyLight 488-labeled goat anti-rat IgM (Thermo Fisher Scientific, Waltham, MA) for HECA-452 for 30 min. After washing, samples were mounted in 50 % glycerol and observed under an immunofluorescence microscope.

2.6. Immunoprecipitation and western blot analysis

Cells were resuspended in phosphate-buffered saline (PBS) supplemented with a protease inhibitor cocktail (Roche, Basel, Switzerland).

Table 1
Clinicopathological information of ALK+ ALCL cases examined in this study.

Case No.	Age (years)	Sex	Location	Origin of sample	CD30	ALK protein localization	PSGL-1	sLe^x	Skin lesion
1	40	Female	Supraclavicular	Lymph node	+	Nucleus and cytoplasm	+	+	–
2	5	Male	Cervical	Mass	+	Nucleus and cytoplasm	+	+	–
3	15	Male	Abdominal	Lymph node	+	Nucleus and cytoplasm	+	+	–
4	9	Male	Abdominal	Lymph node	+	Nucleus and cytoplasm	+	+	–
5	10	Male	Inguinal	Lymph node	+	Cytoplasm only	+	+	+
6	8	Male	Abdominal	Lymph node	+	Nucleus and cytoplasm	+	+	–
7	19	Male	Subcutaneous	Mass	+	Nucleus and cytoplasm	+	+	–
8	59	Female	Cervical	Lymph node	+	Cytoplasm only	+	+	–
9	10	Male	Inguinal	Lymph node	+	Cytoplasm only	+	+	+
10	24	Male	Subcutaneous	Mass	+	Nucleus and cytoplasm	+	+	+
11	75	Female	Cervical	Lymph node	+	Nucleus and cytoplasm	+	+	N/A
12	14	Female	Axillay	Lymph node	+	Cytoplasm only	+	+	+

ALK, anaplastic lymphoma kinase; ALCL, anaplastic large cell lymphoma; PSGL-1, P-selectin glycoprotein ligand 1; sLe^x, sialyl Lewis x

The membrane fraction was obtained as described [24,25] and incubated 15 min with KPL-1 or negative control non-immune mouse IgG (Beckman Coulter, Indianapolis, IN) at 4°C on a rotator. Protein G Sepharose Fast Flow (Sigma-Aldrich, St. Louis, MO) was then added to the membrane fraction/antibody mix, and samples were incubated overnight at 4°C on a rotator. After washing with Tris-buffered saline (TBS), immune complexes were eluted by treating beads with sample buffer, and western blot analysis was conducted as described [26,27]. Briefly, after a 10 min incubation at 75°C, the protein solution obtained by centrifugation (12,000 x *g*, 30 sec) of experimental and control samples was subjected to 4–12 % gradient sodium dodecyl sulfate-polyacrylamide gel electrophoresis and transferred onto a polyvinylidene fluoride membrane (Millipore, Billerica, MA). After blocking 60 min in TBS supplemented with 5 % skim milk, the membrane was incubated overnight at 4°C with KPL-1 or HECA-452 antibodies, followed by incubation with horseradish peroxidase-conjugated anti--mouse IgG or anti-rat IgM antibodies (Jackson ImmunoResearch, West Grove, PA). The membrane was developed using SuperSignal West Dura Extended Duration Substrate (Thermo Fisher Scientific) and evaluated using a luminescent image analyzer LAS-4000 (GE Healthcare, Little Chalfont, UK).

2.7. Bioinformatics analysis

Expression profiles of genes encoding two glycosyltransferases critical for biosynthesis of core 2 *O*-glycan-based sLe^x tetrasaccharide—core 2 β1,3 *N*-acetylglucosaminyltransferase 1 (C2GlcNAcT-1) [28] and α1, 3-fucosyltransferase 7 (FucT-7) [29,30]—in human lymphoma/leukemia cell lines were obtained from the Dependency Map (DepMap) portal (https://depmap.org/portal/). First, we set "Lineage" to "Lymphoid" using the "Cell Line Selector" to obtain 260 cell lines. We then set "Lineage Sub-subtype" to "Anaplastic Large-Cell Lymphoma, ALK--Positive" to define the ALK+ ALCL group (seven cell lines). The remaining 253 lymphoid cell lines were classified into an "other" lymphoma/leukemia group. Gene expression profiles of cell lines in both groups were obtained from the Expression Public 24Q4 dataset. Among all of these lines, data supporting expression of C2GlcNAcT-1 and FucT-7 was seen in seven lines in the ALK+ ALCL group and 179 lines in the other group (186 cell lines in total).

2.8. Reverse transcription quantitative polymerase chain reaction (RT-qPCR)

Total RNA was extracted from cells using ISOGEN reagent (Nippon Gene, Tokyo, Japan), and single-stranded cDNA was synthesized as described [31]. Real-time PCR was performed on a StepOnePlus™ Real-Time PCR System (Applied Biosystems, Waltham, MA) using THUNDERBIRD Next SYBER qPCR Mix (TOYOBO, Osaka, Japan) with the following oligonucleotide pairs: 5′-gTgTCACCTTCCACAgTCCT-3′ and 5′-TCTggTCCCAggAAACCTCA-3′ for C2GlcNAcT-1; 5′-CCACCTATgAggCCTTCgTg-3′ and 5′-AAgCgTTggTATCggCTCTC-3′ for FucT-7; and 5′-TCAAgATCATCAgCAATgCC-3′ and 5′-CgATACCAAAgTTgTCATggA-3′ for glyceraldehyde-6-phosphate dehydrogenase (GAPDH). After initial denaturation at 95ºC for 30 sec, samples were subjected to 40 cycles of amplification with denaturation at 95ºC for 5 sec, annealing and extension at 60ºC for 30 sec. Expression levels of target genes were normalized to those of GAPDH and relative quantification was calculated using the ΔΔCt method against Jurkat cells.

2.9. P-selectin•IgM chimera binding assay

To obtain recombinant P-selectin•IgM chimeric proteins, COS-1 cells (ATCC) were transiently transfected with pcDNA1.1-P-selectin•IgM [23, 32,33] using Lipofectamine Plus (Thermo Fisher Scientific). Transfected cells were cultured 72 hours and then conditioned medium was recovered. P-selectin•IgM binding to cells was assayed by flow cytometry, as described [33].

3. Results

3.1. All 12 cases previously characterized as ALK+ ALCL show histologic features consistent with that diagnosis

We first reviewed tissue specimens from 12 patients previously diagnosed with ALK+ ALCL to reconfirm the diagnosis. Histologically, all specimens exhibited proliferative large cells with abundant cytoplasm and kidney-like or horseshoe-shaped nuclei, or "hallmark" cells (Fig. 1A). Moreover, we observed that lymphoma cells proliferate in (Fig. 1B) and around (Fig. 1C) blood vessels, as well as along lymphatic sinuses (Fig. 1D). Of the 12 cases analyzed, nine (75.0 %) showed a cohesive pattern reminiscent of metastatic carcinoma, five (41.7 %) showed perivascular lesions, seven (58.3 %) showed lymphatic sinus infiltration, and three (16.7 %) showed necrosis with infiltration of various inflammatory cells including eosinophils, neutrophils and plasma cells. As shown in Table 1, skin lesions were observed in four of the 12 cases, all of which were considered to be skin infiltrates from subcutaneous masses. Interestingly, three of the four skin lesions showed a perivascular growth pattern, suggesting that lymphoma cells retain vascular tropism in skin lesions. Immunohistochemically, all cases were CD30-positive (Fig. 1E). We observed ALK protein expression in the nucleus and cytoplasm of lymphoma cells (Fig. 1F) in eight of 12 cases (66.7 %), suggesting the presence of the *NPM1-ALK* fusion gene, while the remaining four cases (33.3 %) showed ALK protein only in cytoplasm (not the nucleus), suggesting the presence of genetic mutations other than *NPM1-ALK* (Table 1). These findings indicate that all 12 cases are typical ALK+ ALCL.

3.2. All 12 ALK+ ALCL cases examined express both PSGL-1 and sLe^x glycans

We next performed immunohistochemical staining for PSGL-1 and sLe^x glycans in all 12 ALK+ ALCL cases described above. As shown in Fig. 1G, in all the plasma membrane of lymphoma cells was diffusely and strongly stained for PSGL-1, and a significant proportion of these cells was also positive for sLe^x glycans (Fig. 1H), consistent with previous reports [12,14]. We also observed some sLe^x-positive lymphoma cells present in the vascular lumen adhering to the luminal surface (Fig. 1I). We then carried out double immunofluorescence staining and observed PSGL-1 co-localization with sLe^x glycans in a significant proportion of lymphoma cells (Fig. 2, top row). We observed similar co-localization in Ki-JK and Karpas 299 cells, which are both ALK+ ALCL lines (Fig. 2, second and third rows, respectively). In contrast, control Jurkat cells expressed PSGL-1 but no sLe^x glycans (Fig. 2, bottom row). These findings suggest that PSGL-1 expressed in ALK+ ALCL is likely decorated with sLe^x glycans.

3.3. PSGL-1 expressed in ALK+ ALCL cells is decorated with sLe^x glycans

To determine whether PSGL-1 expressed in ALK+ ALCL cells is modified with sLe^x glycans, we first performed western blot analysis of membrane fractions of cell lysates prepared from Ki-JK, Karpas 299, and Jurkat lines. PSGL-1 immunoblotting (Fig. 3A, left three lanes) revealed two bands migrating at ~140 kDa and ~280 kDa corresponding to PSGL-1 monomers and homodimers, respectively, in all three lines. On the other hand, sLe^x glycan immunoblotting (Fig. 3A, right three lanes) revealed two bands of the same size as that seen in PSGL-1 immunoblotting only in Ki-JK and Karpas 299 but not in control Jurkat cells. We then immunoprecipitated the membrane fraction of Karpas 299 cells with anti-PSGL-1 monoclonal antibody and subjected precipitates to western blot analysis. As shown in Fig. 3B, two bands whose sizes corresponded to respective PSGL-1 monomers and homodimers were

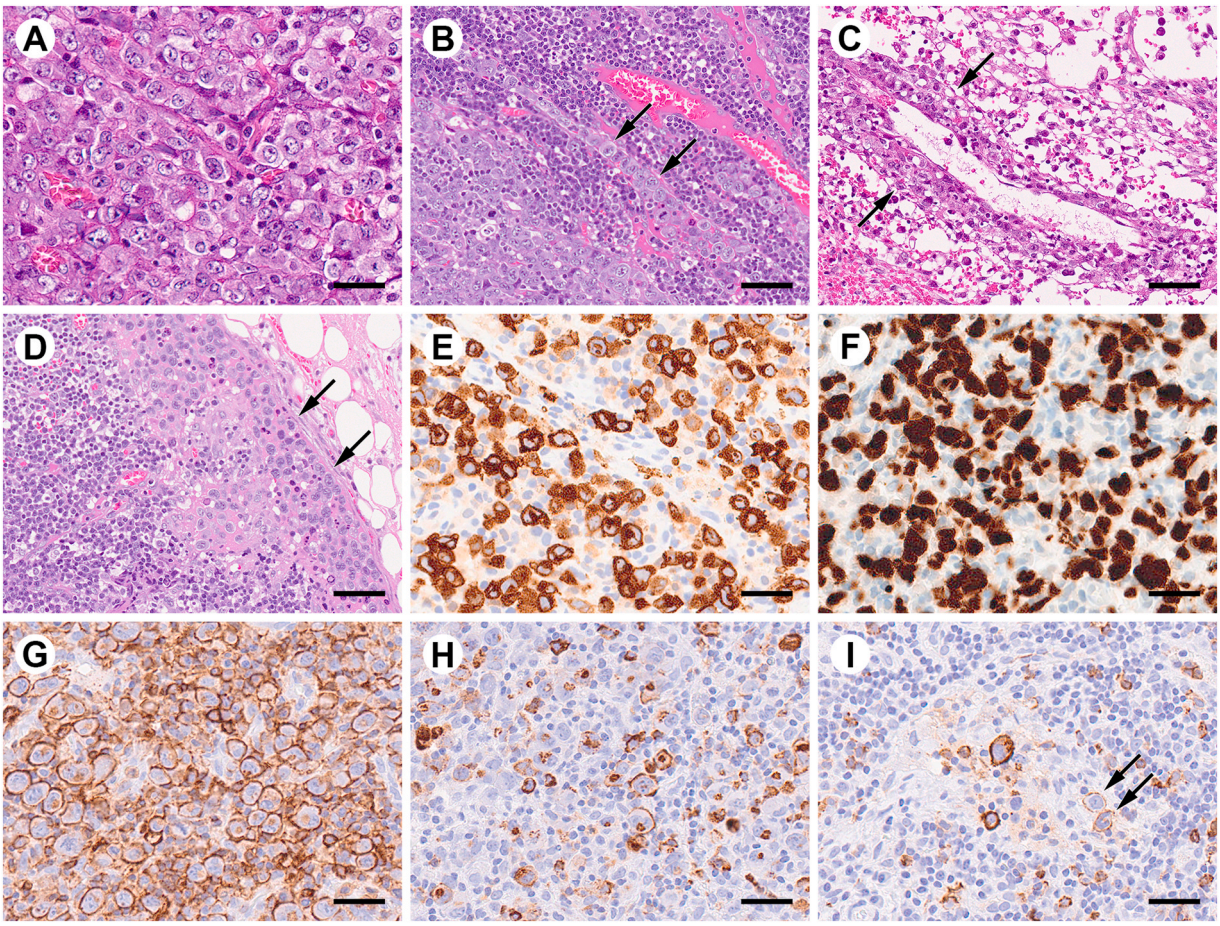


Fig. 1. Histopathological appearance of ALK+ ALCL based on hematoxylin and eosin (H&E) staining (A–D) and immunohistochemical profile (E–I). (A) Undifferentiated lymphoma cells, or "hallmark" cells, proliferate in a cohesive pattern. (B, C, D) Lymphoma cells also proliferate within (arrows in B) and around (arrows in C) blood vessels, and along lymphatic sinuses (arrows in D). (E–I) Tissue sections were immunostained for CD30 (E), ALK (F), PSGL-1 (G) or sLe^x glycans (H and I). Signals were visualized with 3,3'diaminobenzidine (brown) and tissues were counterstained with hematoxylin. Note large lymphoma cells expressing sLe^x glycans attached to the luminal surface of venules in lymph nodes (arrows in I). Bars = 25 μm for panels A and E–I, and 50 μm for panels B–D.

detected in both PSGL-1 and sLe^x glycan immunoblots, but these bands were absent in samples immunoprecipitated with control non-immune mouse IgG. These results clearly indicate that PSGL-1 expressed in ALK+ ALCL cells is decorated with sLe^x glycans.

3.4. sLe^x glycans decorating PSGL-1 are most likely presented on core 2 O-glycans

As noted in the Introduction, a prerequisite for PSGL-1 binding to P-selectin is that sLe^x tetrasaccharide is presented as a capping structure of core 2 *O*-glycans [9]. Formation of core 2 structures, namely, addition of a GlcNAc residue to the *N*-acetylgalactosamine (GalNAc) residue comprising core 1 *O*-glycans (or T antigen) via a β1,6-linkage, is catalyzed by the glycosyltransferase core 2 β1,3 *N*-acetylglucosaminyltransferase 1 (C2GlcNAcT-1) [28]. Given that the DepMap portal (https://depmap.org/portal/) greatly facilitates analysis of expression profiles of specific genes in a vast number of cell lines, we first used it search for expression levels of C2GlcNAcT-1 in ALK+ ALCL and other lymphoma/leukemia cell lines. Consequently, we found that C2GlcNAcT-1 expression is widely observed in various lymphoma/leukemia cell lines, including those derived from ALK+ ALCL (Fig. 4A). On the other hand, expression of α1,3-fucosyltransferase 7 (FucT-7) in ALK+ ALCL cell lines was significantly higher than that observed in other lymphoma/leukemia-derived cell lines (~2^6 or a 64-fold increase in transcripts per million [TPM]) (Fig. 4B). Given that FucT-7 glycosyltransferase adds a fucose residue to the GlcNAc residue comprising sialyl *N*-acetyllactosamine, which is a precursor of sLe^x tetrasaccharide, via an α1,3-linkage [29,30], this finding is consistent with the fact that sLe^x glycans are expressed in Ki-JK and Karpas 299 cells but rarely in Jurkat cells. To confirm these findings, we performed RT-qPCR analysis of transcripts encoding C2GlcNAcT-1 and FucT-7 glycosyltransferases in Ki-JK, Karpas 299 and Jurkat cells. As shown in Fig. 4C, although the relative expression levels of C2GlcNAcT-1 in Ki-JK and Karpas 299 cells are significantly higher than those seen in Jurkat cells, this glycosyltransferase is essentially expressed in all three lines. In contrast, the relative expression levels of FucT-7 in Ki-JK and Karpas 299 cells are dramatically increased ~250- and ~500-fold, respectively, compared to those seen in Jurkat cells. These results combined strongly suggest that sLe^x glycans decorating PSGL-1 in ALK+ ALCL cells are likely presented on core 2 *O*-glycans.

3.5. The PSGL-1 glycoform expressed on ALK+ ALCL cells binds to P-selectin

Finally, to assess functionality of the PSGL-1 glycoform expressed on ALK+ ALCL cells, we performed flow cytometric analysis of Ki-JK, Karpas 299 and Jurkat cell lines using recombinant P-selectin•IgM chimeric protein. As shown in Fig. 5, Ki-JK and Karpas 299 cells, which express both PSGL-1 and sLe^x glycans, were bound by the P-selectin•IgM chimeric protein, while control Jurkat cells, which express PSGL-1 but not sLe^x glycans, were not. This finding indicates that the PSGL-1 glycoform expressed on ALK+ ALCL cells has the capacity to bind to P-

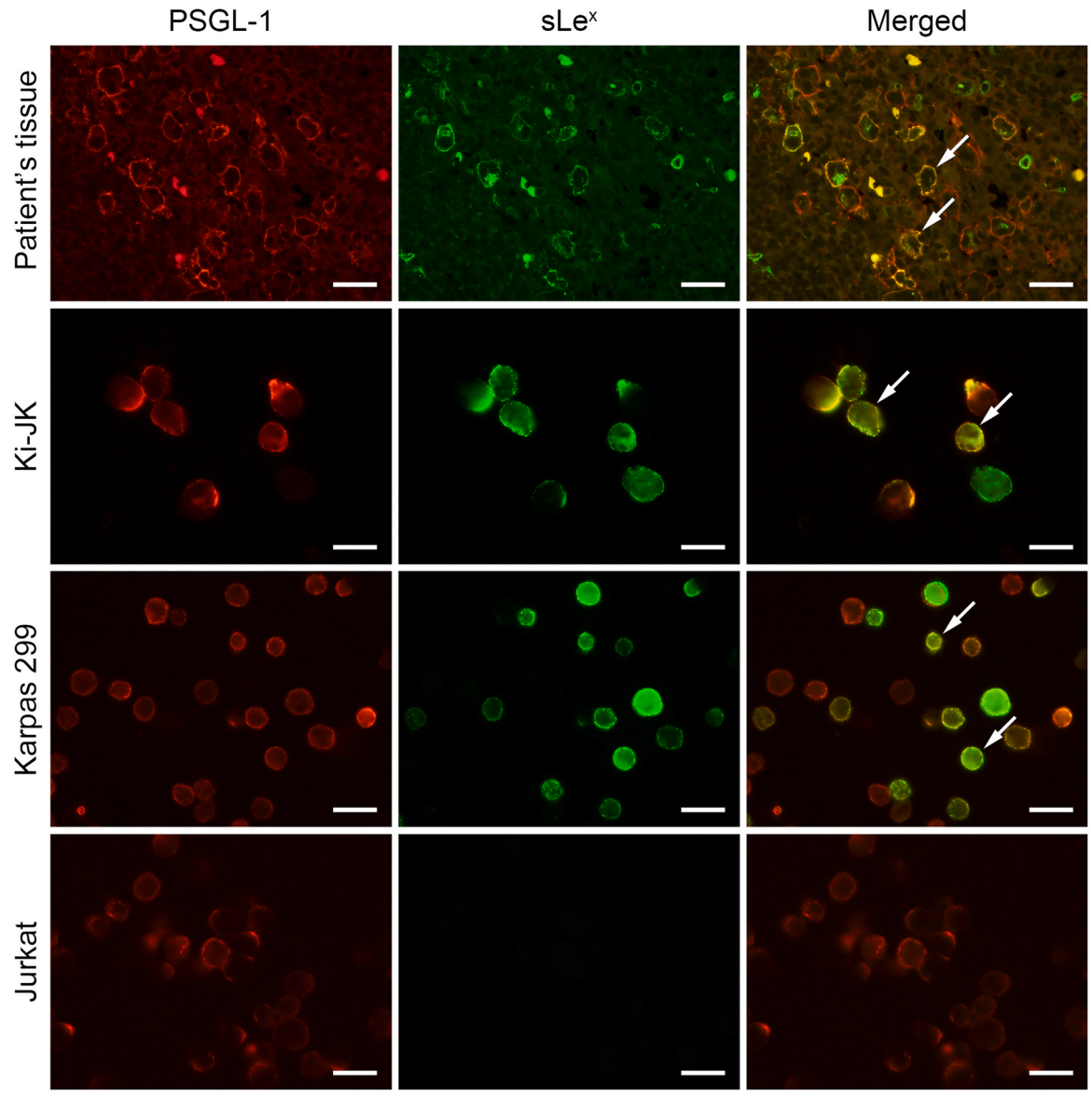


Fig. 2. Double immunofluorescence staining for PSGL-1 and sLe^x glycans. Lymph node tissue sections from a representative ALK+ ALCL patient, two cell lines (Ki-JK and Karpas 299) derived from human ALK+ ALCL specimens, and a control human T-cell leukemia-derived cell line (Jurkat) were doubly immunostained for PSGL-1 (red) and sLe^x glycans (green). Yellow signals in merged images (some indicated by arrows) indicate co-localization of the two antigens. Bar = 25 μm.

selectin.

4. Discussion

In the present study, we show that all 12 confirmed cases of ALK+ ALCL examined, as well as two cell lines derived from ALK+ ALCL patients, express both PSGL-1 and sLe^x glycans. We also show that PSGL-1 expressed on ALK+ ALCL cells is decorated with sLe^x glycans. Given that ALK+ ALCL cells express C2GlcNAcT-1, a core 2-branching glycosyltransferase, we conclude that the sLe^x tetrasaccharide that decorates PSGL-1 is most likely present on core 2 *O*-glycans. Finally, we show that ALK+ ALCL cells are bound by P-selectin•IgM chimeric protein, indicating that the PSGL-1 glycoform expressed on these cells is functional.

In this study, we used the HECA-452 monoclonal antibody to detect sLe^x glycans on PSGL-1. We did this in part because HECA-452 is a rat IgM class antibody, enabling us to conduct double immunofluorescence staining for PSGL-1 and sLe^x glycans using the KPL-1 monoclonal antibody (mouse IgG). HECA-452 also recognizes core 2 *O*-glycan-based sLe^x tetrasaccharide regardless of GlcNAc-6-*O*-sulfation (that is, it recognizes both non-sulfated sLe^x and GlcNAc-6-*O*-sulfated sLe^x [6-sulfo sLe^x]) [18,19,34]. HECA-452 was originally developed to detect unique antigens on high endothelial venules (HEVs) in human secondary lymphoid organs [17], and its epitope was later identified as sLe^x, which overlaps with 6-sulfo sLe^x, which constitutes the carbohydrate moiety of peripheral lymph node addressin (PNAd) [35]. Subsequently, HECA-452 was also found to react with a subset of skin-homing memory T cells [36], and hence to the alias cutaneous lymphocyte antigen (CLA) [37]. Ohmori *et al.* previously proposed that CLA is likely to be 6-sulfo sLe^x [38], but it is not yet known whether CLA is displayed on core 2 *O*-glycans. It will be of interest to determine whether PSGL-1 decorated with core 2 *O*-glycans capped with 6-sulfo sLe^x promotes skin-specific T cell recruitment.

It is well known that ALK+ ALCL often exhibits a cohesive pattern, in which large undifferentiated cells appear to adhere to one another, giving them an epithelial tumor-like appearance [4,5]. This observation led us to hypothesize that, in addition to functionally glycosylated PSGL-1, ALK+ ALCL cells express one of the three selectins (P-, E- or L-selectin) that serve as PSGL-1 ligands and that their adhesive

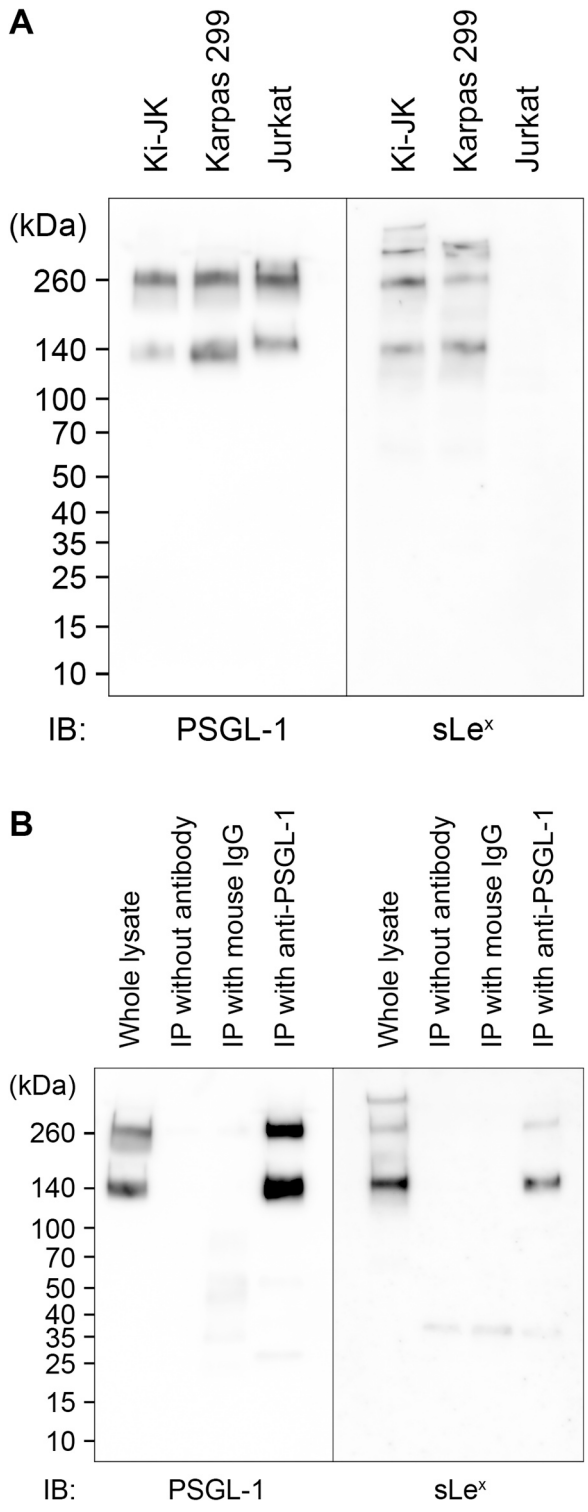


Fig. 3. PSGL-1 expressed in ALK+ ALCL cells is decorated with sLe^x glycans. (A) Western blot analysis of membrane fractions derived from Ki-JK, Karpas 299 or Jurkat cell lysates. (B) Western blot analysis of Karpas 299 cell lysates immunoprecipitated (IP'd) with anti-PSGL-1 antibody. In (A) and (B), membranes were immunoblotted (IB'd) with anti-PSGL-1 monoclonal antibody KPL-1 and anti-sLe^x glycan monoclonal antibody HECA-452. Numbers at left indicate molecular weights (kDa).

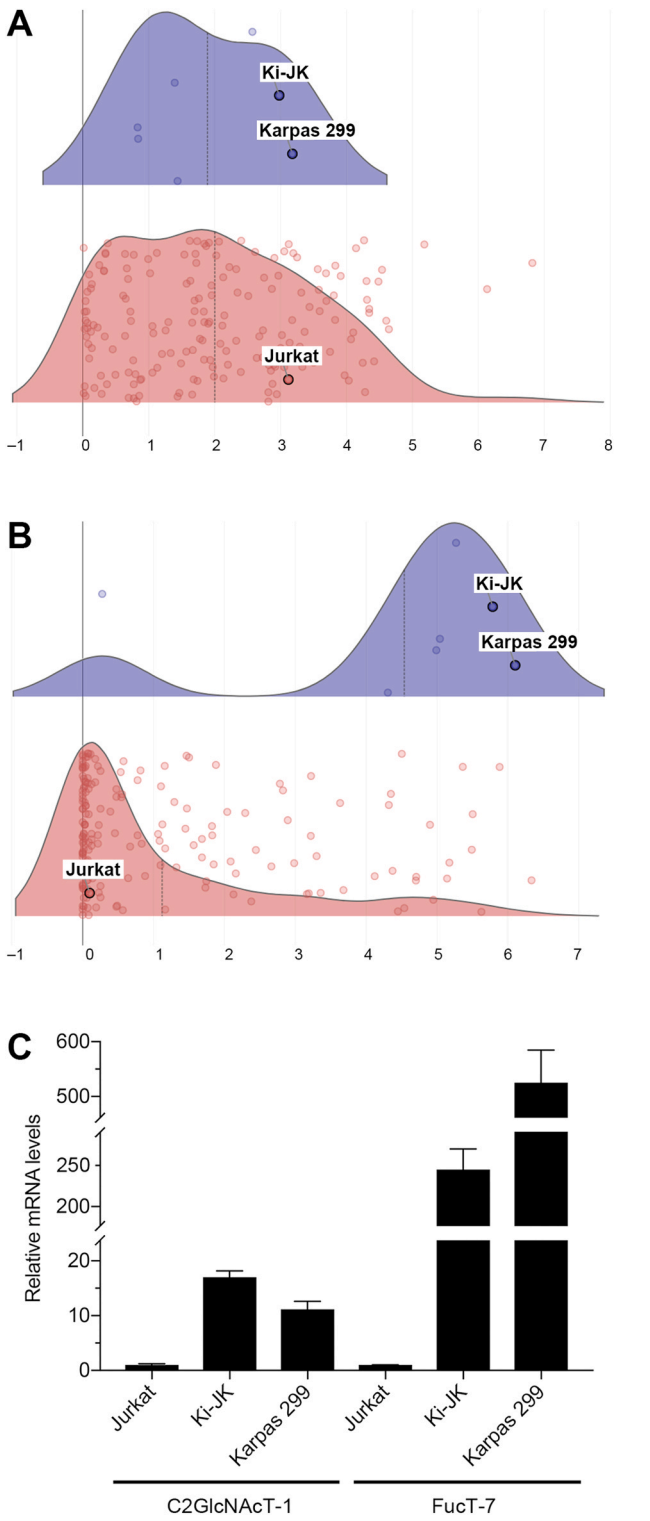


Fig. 4. Expression of transcripts encoding two glycosyltransferases (C2GlcNAcT-1 and FucT-7) critical for functional PSGL-1 glycosylation. (A, B) Expression of C2GlcNAcT-1 (A) and FucT-7 (B) in human lymphoma/leukemia cell lines plotted based on DepMap portal data. Blue and red histograms represent expression in ALK+ ALCL and non-ALK+ ALCL leukemia/lymphoma cells, respectively. X- and Y-axes show logarithm (to the base 2) of transcripts per million (TPM) plus 1, and probability density, respectively. (C) Relative mRNA levels of C2GlcNAcT-1 and FucT-7 in Jurkat (control), Ki-JK and Karpas 299 cells, as determined by RT-qPCR analysis. Data are presented as means with standard deviation (SD) of triplicate measures.

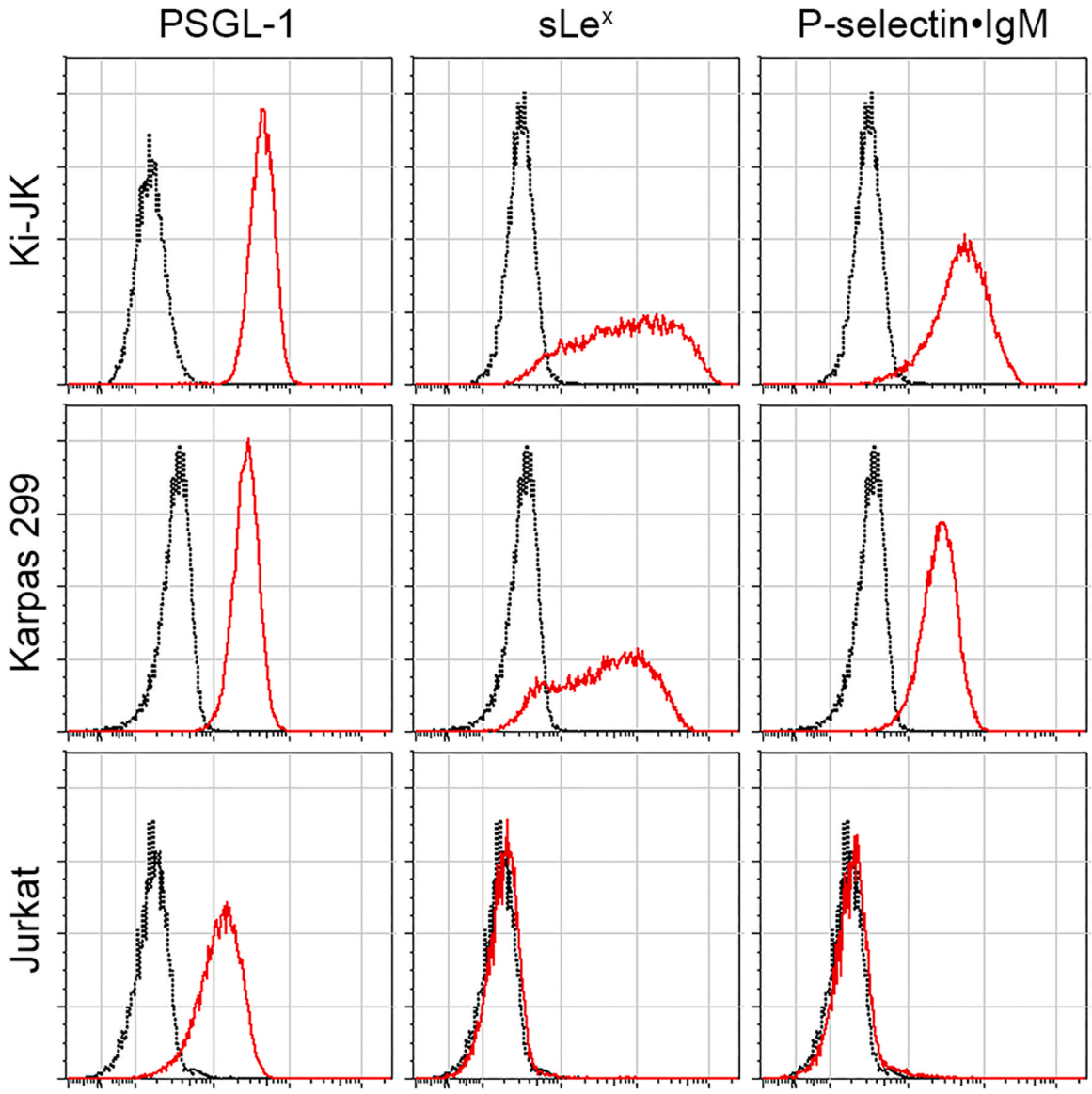


Fig. 5. Binding of P-selectin•IgM chimeric proteins to ALK+ ALCL cells expressing functionally glycosylated PSGL-1. Ki-JK, Karpas 299 and control Jurkat cells were immunostained for PSGL-1 and sLe^x glycans, or incubated with P-selectin•IgM chimeric proteins, and subjected to flow cytometry. Black histograms represent negative controls established by omitting primary antibodies or using conditioned medium of mock-transfected COS-1 cells. X- and Y-axes indicate fluorescence intensity and number of events, respectively.

interactions promote cell-to-cell cohesiveness. To test this hypothesis, we evaluated potential selectin expression in ALK+ ALCL cells using the DepMap portal but found that no selectins were significantly expressed in ALK+ ALCL cells (data not shown). Thus, adhesion molecules other than the PSGL-1–P-selectin axis may function in intercellular cohesiveness seen in ALK+ ALCL. Another possibility is that lymphoma cells proliferate at high density in a confined space (such as the lymphatic sinus) and appear to be attached to one another.

Here we demonstrate that PSGL-1 expressed on ALK+ ALCL cells can bind to P-selectin. Since P-selectin is expressed on endothelial cells comprising HEVs in secondary lymphoid organs and HEV-like vessels in chronic inflammatory sites [39,40], the adhesive interaction between functionally glycosylated PSGL-1 and P-selectin potentially contributes to vascular-related histogenesis of ALK+ ALCL. Indeed, we observed ALK+ ALCL cells attached to the luminal aspect of HEV-like vessels (see arrows in Fig. 1I). Furthermore, given that P-selectin expression is induced in response to various stimuli in virtually all endothelial cells lining venules throughout the body [39,40], it is tempting to speculate that interaction of glycosylated PSGL-1 with P-selectin plays at least a partial role in systemic progression of ALK+ ALCL. Further research is, however, needed to support this hypothesis.

It would be interesting to see whether glycosylated PSGL-1 expressed on ALK+ ALCL also binds E-selectin. Mitoma *et al.* previously demonstrated that Chinese hamster ovary (CHO) cells expressing PSGL-1 decorated with core 2 *O*-glycan-based sLe^x tetrasaccharide bound not only P-selectin but also L- and E-selectin [33]. However, these authors concentrated the soluble E-selectin protein ~10 times to obtain binding equivalent to P-selectin [33]. This finding is consistent with the report by Moore *et al.* demonstrating that glycosylated PSGL-1 binds to E-selectin, but with at least 50-fold lower affinity than P-selectin [7]. Taken together, these reports strongly suggest that glycosylated PSGL-1 expressed on ALK+ ALCL cells may bind weakly to E-selectin. Further studies are, however, needed to clarify this issue.

In this study, we showed that while three cell lines tested (Ki-JK, Karpas 299 and Jurkat cells) express PSGL-1, ALK+ ALCL cells, but not T-cell leukemia cells, express sLe^x glycans to form functional PSGL-1. This result is in agreement with previous reports that PSGL-1 is constitutively expressed in a wide range of T cells, but that resting T cells

(naive and central memory T cells) deficient in C2GlcNAcT activity express a non-selectin-binding form of PSGL-1 in which Thr57 is not appropriately glycosylated [41], which is a requirement for P-selectin binding but not chemokine engagement [10]. Thus, P-selectin-binding capacity of PSGL-1 is post-translationally regulated by glycosylation of PSGL-1 with sLe^x glycans, which enable effective immune cell trafficking.

In addition to functioning as a cell adhesion molecule, PSGL-1 was recently found to be an immune checkpoint protein that negatively regulates T cell responses in a manner that promotes T cell exhaustion and induces immune tolerance [42]. Thus, PSGL-1 is anticipated to become a new target for immune checkpoint inhibitors in T-cell lymphomas [43]. Whether the presence or absence of PSGL-1 glycosylation affects therapeutic efficacy awaits further studies.

5. Conclusion

ALK+ ALCL expresses PSGL-1 decorated with core 2 *O*-glycans terminated with sLe^x tetrasaccharide. This PSGL-1 glycoform is functional and capable of binding to P-selectin expressed on vascular endothelium. The PSGL-1–P-selectin axis may contribute to the vascular-related histogenesis characteristics of ALK+ ALCL.

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CRediT authorship contribution statement

Ohtani Haruo: Resources, Project administration. **Komeno Takuya:** Resources. **Yonemoto Natsumi:** Writing – original draft, Project administration, Investigation, Formal analysis, Data curation. **Kobayashi Motohiro:** Writing – original draft, Supervision, Project administration, Funding acquisition, Formal analysis. **Hoshino Hitomi:** Investigation. **Fukushima Mana:** Supervision, Conceptualization. **Muramoto Akifumi:** Writing – review & editing, Formal analysis. **Fukiage Yusuke:** Investigation. **Nakanuma Yasuni:** Resources. **Kaizaki Yasuharu:** Resources. **Inadome Yukinori:** Resources.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Motohiro Kobayashi reports financial support was provided by Japan Society for the Promotion of Science. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.prp.2025.155988.

Data Availability

Data supporting findings reported in this study are available from the corresponding author upon reasonable request.

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